

The essence of Newton's method was to identify the forces acting on an object, to predict the course of its motion with mathematical rigour and check that prediction against direct observation. The underlying assumption was that mechanical systems behave in definitely predictable ways. When it came to molecules the rules of the game suddenly changed. What troubled Maxwell was that molecules were too small to be seen, too numerous to be accounted for individually and too quick to be captured. He had lost the luxury of certainty – turned to probability: More important by replacing Electromagnetic forces by fields, cleared the way for the scientific revolutions of the Twentieth century. In 1925 Werner Heisenberg postulated that the electrons occupy different regions in the vicinity of atomic nucleus with different probabilities. It was a revolutionary way to think but yielded spectacularly accurate results. Just as Maxwell brought probability into Heat, Heisenberg introduced probability into the realm of atomic physics.

von Baeyer

1. High School review. Below $n \geq 1$, and $0 \leq k \leq n$ integers. Show

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$$

$$\binom{n}{0}\binom{m}{k} + \binom{n}{1}\binom{m}{k-1} + \binom{n}{2}\binom{m}{k-2} + \cdots + \binom{n}{k}\binom{m}{0} = \binom{n+m}{k}$$

$$\binom{n}{k} - \binom{n}{k-1} + \cdots \mp \binom{n}{1} \pm 1 = \binom{n-1}{k}$$

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2 = \binom{2n}{n}$$

$$\sum_{j=0}^n \frac{(2n)!}{[j!]^2[(n-j)!]^2} = \binom{2n}{n}^2$$

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots = n2^{n-1}$$

$$\binom{n}{0}\binom{n}{k} - \binom{n}{1}\binom{n-1}{k-1} + \binom{n}{2}\binom{n-2}{k-2} - \cdots \pm \binom{n}{k}\binom{n-k}{0} = 0$$

Use $\sum_{j=0}^{n-1} (1-t)^j = [1 - (1-t)^n]t^{-1}$ to show

$$\binom{n}{1}\frac{1}{1} - \binom{n}{2}\frac{1}{2} + \binom{n}{3}\frac{1}{3} - \cdots + (-1)^{n-1}\binom{n}{n}\frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

Show $\sum_{k=0}^n (-1)^{n-k} \binom{n}{k} k^r = 0$ or $n!$ according as $r < n$ or $r = n$.
 (i) by induction and (ii) using r -th derivative of $(1 - e^t)^n$ at $t = 0$.

2. A poker hand means a set of five cards selected at random from usual deck of playing cards.

(a) Find the probability that it is a Royal Flush - consists of ten, jack, queen, king, ace of one suit. (b) Find the probability that it is four of a kind - there are four cards of equal face value. (c) Find the probability that it is a full house - consists of one pair and one triple of cards with equal face values. (d) Find the probability that it is a straight - consists of five cards in a sequence regardless of suit. (e) Find the probability that it consists of three cards of equal face value and two other cards but not a full house. (f) Find the probability that it consists of two distinct pairs and another card but does not fall into previous categories. (g) Find the probability that it consists of a pair and three other cards but does not fall into previous categories.

3. In how many ways can eight rooks be placed on a chess board so that none can take another and none is on the white diagonal.
4. One mapping is selected at random from the collection of all mappings of $A = \{1, 2, \dots, n\}$ to itself. What is the probability that the selected map transforms each of the n elements into 1?

Let $1 \leq i, k \leq n$ be fixed. What is the probability that element i has exactly k pre-images? What is the probability that the element i is mapped to k ?

Suppose distinct elements $i_1, i_2, i_3 \in A$ are given. Also $j_1, j_2, j_3 \in A$ are given. What is the probability that the elements i_1, i_2 and i_3 are transformed to j_1, j_2 and j_3 respectively?

5. One permutation is selected at random from the set of all permutations of $A = \{1, 2, \dots, n\}$. What is the probability that the identity permutation is chosen?

Assume that distinct points $i_1, \dots, i_k \in A$ are given and also $j_1, \dots, j_k \in A$ are given. What is the probability that the selected permutation transforms $i_1, i_2, \dots, i_k$ to $j_1, j_2, \dots, j_k$ respectively?

Let $i \in A$ be given. What is the probability that the permutation keeps i fixed?

What is the probability that the elements 1, 2, and 3 form a cycle in that order? in some order?

What is the probability that all the elements of A form one cycle?

6. (Maxwell-Boltzman expt; n boxes, r balls)

Show: the probability that box 1 contains exactly k balls is given by $p_k = \binom{r}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{r-k}$.

show that the most probable number of balls in box 1 is given by k where $\frac{r+1}{n} - 1 < k \leq \frac{r+1}{n}$; that is

$$p_0 < p_1 < \cdots < p_{k-1} \leq p_k > p_{k+1} > \cdots > p_r.$$

Let m be fixed. As $n \rightarrow \infty$ and $r \rightarrow \infty$ in such a way that $\frac{r}{n} \rightarrow \lambda$ [do you understand this] show that; $p_m \rightarrow e^{-\lambda} \frac{\lambda^m}{m!}$.

Show that the probability of finding exactly m boxes empty equals

$$p_m(r, n) = \frac{1}{n^r} \binom{n}{m} \sum_{\nu=0}^{n-m} (-1)^\nu \binom{n-m}{\nu} (n-m-\nu)^r.$$

7. (Bose-Einstein expt; n boxes, r balls)

Show the probability that box 1 contains exactly k balls is given by $q_k = \binom{n+r-k-2}{r-k} / \binom{n+r-1}{r}$

Let $n > 2$. Show that zero is the most probable number of balls in box 1. In fact, $q_0 > q_1 > q_2 > \cdots$.

As $n \rightarrow \infty$ and $r \rightarrow \infty$ in such a way that $\frac{r}{n} \rightarrow \lambda$ show, for each m ; $q_m \rightarrow \frac{\lambda^m}{(1+\lambda)^{m+1}}$.

Show: $P(\text{exactly } m \text{ boxes are empty}) = \binom{n}{m} \binom{r-1}{n-m-1} / \binom{n+r-1}{r}$.

8. Let $A_1, \dots, A_N$ be events in an experiment. $1 \leq m \leq N$. let E be the set of all outcomes ω which belong to exactly m of these events, that is, $|\{1 \leq i \leq N : \omega \in A_i\}| = m$. Show

$$P(E) = S_m - \binom{m+1}{m} S_{m+1} + \binom{m+2}{m} S_{m+2} - \cdots \pm \binom{N}{m} S_N.$$

In N card matching, the $P(\text{exactly } m \text{ matches})$ equals

$$p_{[m]} = \frac{1}{m!} \left[1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \cdots \pm \frac{1}{(N-m-1)!} \mp \frac{1}{(N-m)!} \right].$$

9. Two similar decks of N distinct cards are matched against one similar target deck. Show: probability of exactly m double matches is

$$u_m = \frac{1}{m!} \frac{1}{N!} \sum_{k=0}^{N-m} (-1)^k \frac{(N-m-k)!}{k!}.$$

Show: $u_0 \rightarrow 1$ and for each $m \geq 1$, $u_m \rightarrow 0$ as $N \rightarrow \infty$.

This is a story about four people named Everybody, Somebody, Anybody and Nobody. There was an important job to be done. Everybody was asked to do it. Everybody was sure that Somebody would do it. Anybody could have done it, but Nobody did it. Somebody got angry about it because it was Everybody's job. Everybody thought Anybody could do it but Nobody realized that Everybody would not do it. It ended up that Everybody blamed Somebody when Nobody did what Anybody could have done.

Charles Osgood (?)

10. Find the probability that, in five tosses, a coin falls heads at least three times in succession.
Find the probability for a head run of at least length five in ten tosses. In a sequence of symbols H, T ; a run is a maximal sequence of a symbol uninterrupted by the other symbol. For example: $TTTTHHHTHTT$ has 5 runs, two H runs of lengths 3 and 1; three T runs of lengths 4, 1, 2.
11. A tournament (directed graph, exactly one directed edge between any two vertices. $v \rightarrow w$, v wins over w) is said to have k -leader property if for every set of k players there is one who beats them all. Consider a random tournament with n players. Fix k players.
Show: $P(\text{no player beats all these } k) = (1 - 2^{-k})^{n-k}$.
Show k -leader tournaments are possible for all large n .
12. Die A has four red faces and two white faces. Die B has two red and four white faces. A fair coin is flipped once; if Heads we use die A forever; if Tails we use die B forever.
Show probability of red at any throw is $1/2$.
If first two throws resulted in Red, find the chances of red at third throw? If the first n throws resulted in Red, find the chances that die A is being used?
13. A closet contains n pairs of shoes. $2r$ shoes are chosen at random. What are the chances that there is no complete pair? exactly one complete pair? exactly two complete pairs?
14. Picking card from usual deck with replacement: How many drawings are needed so that chances of obtaining at least one ace is at least $1/2$.
15. Matching N envelopes and letters: What are the chances that there is match at place i given that there is no match at position j .

16. Have two urns each having 20 balls numbered $1, 2, \dots, 20$. A set of six balls are drawn from each urn. What is the probability that exactly k numbers are common for both samples ($0 \leq k \leq 6$)? What if the six balls are drawn without replacement from each urn.
17. 10 balls are put in 10 boxes (Maxwell-Boltzman). Given that box number one is empty, what are the chances that only one box is empty? Given that only one box is empty, what are the chances that box one is empty?
18. A fair die is rolled two times. Given that 5 showed at least once what is the probability that it showed both times? exactly once?
19. Using a machine which produces independent random bits; devise an algorithm to pick a number at random from $\{1, 2, 3, 4\}$; same to pick at random from $\{1, 2, 3\}$.
20. Find the chances that a random permutation of $\{1, 2, 3, 4\}$ has exactly two cycles each of length 2? has at least one cycle of length 2? has at least one cycle of length at least two?
21. Polya scheme of adding c balls: Let $1 \leq m < n$ be integers and A, B each stand for any of red/green. Show (time symmetry)
 $P(A \text{ at draw } m | B \text{ at draw } n) = P(A \text{ at draw } n | B \text{ at draw } m)$.
22. Let $p_k(n)$ be chances of exactly k green balls in the first n draws in Polya scheme. interpret $p_{-1}(n) = 0$. Show

$$p_k(n+1) = p_k(n) \frac{r + (n-k)c}{g + r + nc} + p_{k-1}(n) \frac{g + (k-1)c}{g + r + nc}.$$

$$p_k(n) = \binom{-g/c}{k} \binom{-r/c}{n-k} \bigg/ \binom{-(r+g)/c}{n} \quad 0 \leq k \leq n.$$

Notation: $\binom{a}{k} = \frac{a(a-1)\dots(a-k+1)}{1 \cdot 2 \dots k}$ for $k \geq 1$ an integer and $\binom{a}{0} = 1$.

23. r alphas and s betas are arranged in a row, at random. Show: the chances that the arrangement contains a total of k runs equals

$$2 \binom{r-1}{\nu-1} \binom{s-1}{\nu-1} \bigg/ \binom{r+s}{r} \quad k = 2\nu;$$

$$\left\{ \binom{r-1}{\nu} \binom{s-1}{\nu-1} + \binom{r-1}{\nu-1} \binom{s-1}{\nu} \right\} \bigg/ \binom{r+s}{r} \quad k = 2\nu + 1.$$

24. A student is taking a multiple choice exam in which each question has 4 suggested answers with exactly one correct. the student knows the correct answer for sixty percent of the questions. If he knows the answer he chooses the correct answer, otherwise he selects at random. What is the probability that for a given question he selects the correct answer? Given that he chose the correct answer what are chances that he guessed?
25. Let $A_1, \dots, A_9$ be independent.
- Show that $A_1 \cup A_2; A_3 \cap A_4, A_5^c \cup A_6^c, A_7^c, A_8 \cup A_9^c$ are independent.
 - show $P\{(A_1 \cup A_5) \cap (A_7^c \cup A_8) | (A_2 \cup A_3^c)\} = P\{(A_1 \cup A_5) \cap (A_7^c \cup A_8)\}$.
26. I roll a fair die n times (independently). Let A_i be the event that i -th throw is even; for $1 \leq i \leq n$. Let A_{n+1} be the event that the total sum is even. Show that any n of these $(n+1)$ events are independent. Show that the $(n+1)$ events are not independent.
27. Kshitiz and Kabir are equally good in Badminton, they play a series of games (independent and no draws) till someone wins 25 games and stop. What are the chances that when the match ends the looser won 21 games.? won 24 games?
28. Want to see if two polynomials $P(x), Q(x)$ are same. Their degrees are at most d . I can only access their values at any given point, but only once. Here is an algorithm: Pick a number r at random from $\{1, 2, \dots, 100d\}$. If $P(r) = Q(r)$ declare P, Q are same; otherwise not same. Show you are right with chance 0.99.
29. (i) I am trying to send you a single bit, either a 0 or a 1. When I transmit the bit, it goes through a series of n relays before it arrives to you. Each relay flips the bit independently with probability p . Show that the chances of your receiving correct bit is $\sum_0^{\lfloor n/2 \rfloor} \binom{n}{2k} p^{2k} (1-p)^{n-2k}$.
- Let us say that a relay has bias q if it flips the bit with probability $(1-q)/2$. Thus $q \in [-1, 1]$. Show that sending a bit through relays with biases q_1 and q_2 is equivalent to sending through a single relay with bias $q_1 q_2$.
 - Prove that probability of receiving correct bit when it passes through n relays as in part (i) is $\frac{1+(1-2p)^n}{2}$.

(most problems are from Feller and Mitzenmacher).

Richard Feynman hated math class. The problem was that teacher always gave you, right alongside the question to solve, the method by which to solve it. Where is the adventure in that? The endeavour felt plodding and bloodless - a government of the dullards by the dullards and for the dullards. Math club he loved. It was a play ground, a school of trickster witchcraft and improvisational wizardry. The problems required only algebra but each possessed a devious twist. If you tried to apply a standard method, you would run out of time. Instead you had to discover a simplifying shortcut.

Ben Orlin

30. For a die, face i appearing in a throw has chance p_i for $1 \leq i \leq 6$. Here $p_i > 0$ for all i and $\sum p_i = 1$. I roll it 100 times independently. Let X be the number of throws when even face appears. Find the distribution of X . Let Y be the number of times when a face other than 5 appears. Find the distribution of Y . Calculate means and variances of X and Y .
31. Throw a die twice independently. Let X and Y be the numbers in the first and second throw. Calculate $E(X \vee Y) - \max$; and $E(X \wedge Y) - \min$. Verify. $E(X \vee Y) + E(X \wedge Y) = E(X) + E(Y)$
32. Give examples of events A, B where $P(A|B) > P(A)$; where $P(A|B) = P(A)$; where $P(A|B) < P(A)$.
Suppose $C_1, C_2, \dots, C_n$ is a partition of the sample space. Let A be an event. If $P(A|C_i) \leq a$ for all i , then show $P(A) \leq a$. If $P(A|C_i) = a$ for all i , then show $P(A) = a$.
33. In this exercise matrices have zero-one entries; arithmetic operations are mod 2. Let A, B, C be $n \times n$ matrices and $AB \neq C$. Show that if you select a vector $v \in \{0, 1\}^n$ at random, then $P(ABv = Cv) \leq 1/2$.
34. Two subsets X, Y of $\{1, 2, \dots, n\}$ are selected independently and at random from among all subsets. Find $P(X \subset Y)$. Find $P(X \cup Y = \{1, 2, \dots, n\})$.
35. We draw cards at random with replacement from usual deck of cards. What is expected number of draws needed till we see all the 52 cards?
36. For a random permutation on $\{1, 2, \dots, n\}$, what is the expected number of fixed points? What is the expected number of cycles?

37. I throw a fair die. If 4,5 or 6 shows up I put X as that value. If 1,2 or 3 shows up I roll the die again and then put X as the sum of the two faces obtained. calculate the distribution of X .
38. A bag contains balls numbered one to hundred. A sample of size r is taken with replacement. Let X be the maximum of the numbers observed. Find the distribution of X for $r = 1, 2, 3, 4$. Find $E(X)$ for these values of r . What if X were minimum.
39. What is the most probable value of a $B(n, p)$ variable?
What is the most probable value of a $P(\lambda)$ variable.
40. I have 100 books. I sample with replacement till I obtain seven different books and then stop. Let N be the size of the sample. Find $E(N)$.
41. There are 100 persons from each of 15 different states (total 1500). I take a sample of size 100 with replacement and N is the number of states not represented in the sample. Find $E(N)$.
42. If $X \sim P(\lambda)$, show $E[X(X-1)\cdots(X-r+1)] = \lambda^r$.
if $X \sim G(p)$, show $E[X(X-1)\cdots(X-r+1)] = \frac{r!(1-p)^r}{p^r}$.
43. If $X \sim G(p)$, show $P(X \geq i+j | X \geq i) = P(X \geq j)$.
Show $E\left(\frac{1}{1+X}\right) = \log\left[p^{p/(p-1)}\right]$.
44. Fix $0 < p < 1$. Then $G(n, p)$ be Erdos-Renyi random graph Let a_n be the probability that the graph is disconnected. Show

$$a_n \leq \frac{1}{2} \sum_{k=1}^{n-1} \binom{n}{k} (1-p)^{k(n-k)} \leq \sum_{k=1}^{\lfloor n/2 \rfloor} \{n(1-p)^{n/2}\}^k \rightarrow 0$$

Thus $1 - a_n \rightarrow 1$. Thus as n becomes large, our random graph is connected with an overwhelming probability. (Analysis at work!)

45. Let $G = (V, E)$ be a graph (**given to you**). Choose points of V at random, each with probability $1/2$. Let T be the random set so obtained. Call an edge 'crossing edge' if one end is in T and the other in $V - T$. Since T is random, the number X of crossing edges is a random variable. Show that $E(X) = e/2$ where e is the total number of edges in the graph, that is, cardinality of E .
Deduce that a graph with n vertices and e edges must have a bipartite subgraph with at least $e/2$ edges. A subgraph of (V, E) means a graph

(V^*, E^*) where $V^* \subset V$; $E^* \subset E$. Carefully note E^* need not be all of E restricted to $V^* \times V^*$. A graph (V, E) is bipartite if $V = V_1 \cup V_2$ with (i) $V_1 \neq \emptyset$ and $V_2 \neq \emptyset$; (ii) $V_1 \cap V_2 = \emptyset$ and (iii) every edge has one vertex in V_1 and other in V_2 .

46. Consider $G(n, p)$ Erdos-Renyi random graph. Say that a pair of vertices (unordered) is good, if either they are joined or they are joined to a common vertex. Let X be the number of pairs which are NOT good. Since the graph is random, X is a random variable. Show that

$$E(X) \leq \binom{n}{2}(1 - p^2)^{n-2} \rightarrow 0$$

Deduce that $P(X \geq 1) \rightarrow 0$. Conclude, with overwhelming probability a $G(n, p)$ random graph has diameter 2; as n becomes large. Diameter of a graph is the largest possible distance between pairs of vertices and distance between a pair of vertices is the number of edges in the shortest path joining them — ∞ if no such path.

47. Show that there is a tournament on n players which has at least $n!2^{-(n-1)}$ Hamiltonian paths. A Hamiltonian path is an ordering of the vertices $v_1, v_2, \dots, v_n$ such that for each $i < n$, $\overrightarrow{v_i v_{i+1}}$ is an edge. Do not confuse with Hamiltonian Cycle.
48. Let $v_1, \dots, v_n \in R^n$ with $\|v_i\| = 1$ for all i . Show that there exist numbers ϵ_i ($1 \leq i \leq n$) each ± 1 such that $\|\sum \epsilon_i v_i\| \leq \sqrt{n}$. Also show: there are numbers η_i ($1 \leq i \leq n$) each ± 1 such that $\|\sum \eta_i v_i\| \geq \sqrt{n}$.
49. Suppose that a store buys b (integer) items in anticipation of a random demand Y , non-negative integer valued. Suppose that each item sold brings a profit of Rs π while each item stocked but unsold brings a loss of Rs λ . The problem is to choose b so as to maximize the expected profit. Show that the least integer k satisfying $P(Y \leq k) \geq \pi / (\lambda + \pi)$ is the best choice of b . What if $\pi = \lambda$? What if $\pi = 3\lambda$ or $\lambda = 3\pi$?
50. Let X be any discrete random variable. Define $F(a) = P(X \leq a)$ for $a \in R$. This function is called the Distribution Function (DF) of X .
- (i) Show that F is monotone non-decreasing; $0 \leq F(a) \leq 1$.
 $\lim_{a \rightarrow -\infty} F(a) = 0$ and $\lim_{a \rightarrow \infty} F(a) = 1$.
 - (ii) Show that F is right continuous and has left limit at every $a \in R$.
 - (iii) Show $P\{X = a\} = F(a) - F(a-)$. Deduce that $P\{X = a\} = 0$ iff F is continuous at a . (Analysis at work!)

One of our most painful feelings is regret. It originates from our not having understood, in the past, how something would have been important to us in the present — and thus not having done what needed to be done to obtain it.

Having discipline with your body, exercising and eating healthily, frees you from the pressure of age and diseases. Having discipline with your work, doing the important things which have to be done, brings you the success that will free you from pressure of studies. Having discipline with your mind frees you from the pressure of your emotions.

It is your responsibility to find out which mental patterns are preventing you from addressing your problems and to find a way to get rid of such mental patterns.

Luca Dellanna

51. X non-negative integer valued, show $E(X) = \sum_1^{\infty} P(X \geq n)$. Rigorously Justify your steps.
52. Consider placing 3 balls in 3 boxes (Maxwell-Boltzman). Let N be the number of occupied boxes. Let X_i be the number of balls in box i for $i = 1, 2, 3$. Calculate joint distribution of (N, X_3) ; of (X_1, X_2, X_3) .
53. In a sequence of Bernoulli trials, let X be the length of the run started by the first trial. Find its distribution, mean, and variance. Let Y be the length of the second run. Find its distribution, mean and variance. Find joint distribution of X, Y .
54. Let X, Y, Z be independent each $G(p)$. Calculate $P(X = Y)$; $P(X \geq 2Y)$; $P(X + Y \leq Z)$.
55. Two dice are thrown. let X be the score on the first die and Y be the larger of the two scores. Find joint distribution of X, Y ; their means, variances and covariance.
56. In five tosses of a fair coin, let X, Y, Z be, respectively, number of Heads; number of Head runs; length of largest Head run. Calculate their joint distribution. Make bivariate tables for each pair. calculate marginals. Find means variances and covariances.
57. We place n bar magnets end to end in independent random orientation. Like ends attract and unlike ends repel making the system into certain

random number of blocks. Find expectation and variance of the number of blocks.

58. $\{X_k\}$ be iid (means; independent and all have the same distribution) strictly positive discrete rvs on (Ω, P) with both $E(X)$ and $E(\frac{1}{X})$ finite. Denote $E(X) = a$. Let $S_n = \sum_1^n X_k$.

Show $E(\frac{1}{S_n})$ exists (yes, need to show) and $E(\frac{X_k}{S_n}) = \frac{1}{n}$ for $1 \leq k \leq n$.

Show $E(\frac{S_m}{S_n}) = \frac{m}{n}$ if $m \leq n$.

Show $E(\frac{S_m}{S_n}) = 1 + (m - n)aE(\frac{1}{S_n})$ if $m \geq n$.

59. let S_n be the number of matches with n cards (or letters-envelopes). We saw $E(S_n) = 1$ no matter what n is. Show $Var(S_n) = 1$.

60. A subset A of integers is said to be sum-free, if sum of two things in A is not in A . More precisely, if you **can not** find $a_1, a_2, a_3 \in A$ such that $a_1 + a_2 = a_3$. We did not say that these a_i be distinct. If $0 \in A$, A is not sum free. if $a, 2a \in A$ then A is not sum-free.

Given a set B of non-zero integers, we can find a sum-free subset $A \subset B$ having at least one-third of the number of elements of B . Here is how (See the power of probability!).

Say, $B = \{b_1, \dots, b_n\}$. Pick a prime $p > 2 \max_{1 \leq i \leq n} |b_i|$ which is of the form $p = 3k + 2$. Show that such a p exists.

Show $C = \{k + 1, k + 2, \dots, 2k + 1\}$ is sum-free in $Z_p = \{0, 1, \dots, p - 1\}$ where addition is mod p .

Pick x at random from the set of non-zero elements of Z_p and set $d_i = x \cdot b_i \text{ mod } (p)$. Show that, for fixed i , as x ranges over the non-zero elements of Z_p , so does d_i . Conclude that $P(d_i \in C) \geq 1/3$.

Show that the expected number of b_i such that $d_i \in C$ is at least $n/3$.

Conclude that there is x , $1 \leq x \leq p - 1$ and an $A \subset B$ with $|A| \geq n/3$ such that $x \cdot a \in C$ for all $a \in A$. Here $x \cdot a$ is taken mod (p) .

Show that A is sum-free.

61. A bakery makes 80 loaves of bread daily of which ten are underweight. An inspector weighs five loaves at random. What is the probability that an underweight loaf will be discovered? Simplify using calculator. (We shall NOT have such problems, needing calculator, in exam.)

62. I have a coin whose chance of heads is p . I toss it $n + 1$ times. For $1 \leq i \leq n$, let X_i be one if both i -th and $(i + 1)$ -th tosses yield heads; and zero otherwise. find the mean and variance of the sum $\sum X_i$.

63. Consider a die with k faces, face i turning up on a throw is p_i for $1 \leq i \leq k$ where $\sum p_i = 1$ and $p_i > 0$ for each i . Roll the die n times independently. Let X_i be the number of times face i appears. Find the joint distribution of $(X_1, X_2, \dots, X_k)$. This is called multinomial distribution: $M(n; p_1, \dots, p_k)$. If you denote $X = (X_1, X_2, \dots, X_k)$, one says $X \sim M(n, p_1, \dots, p_k)$. What is the marginal distribution of X_5 . What is the distribution of (X_1, X_2, X_3) . What is the distribution of (X_1, X_3, X_5, X_7) What is the distribution of $X_1 + X_3 + X_5$.
64. Toss a coin; chance of heads p ; till you get r heads and stop. Let Z be number of tails. Define X_1 : number of T before first H ; X_2 : number of T between first and second H ; etc. Show $X_i; 1 \leq i \leq r$ are independent. Calculate mean and variance of $NB(r, p)$ variable.
65. Suppose that there are 30 people in a room and assume that their birthdays are a random selection, with replacement, from among the 365 days. Find the chances that they are distinct. Use calculator and evaluate for 30 people, for 35, for 40. You see that with high chance two birthdays coincide. People believed that since there are 365 possibilities this is unbelievable. This is called birthday paradox. Since for large n , $(1 - \frac{j}{n}) \sim e^{-j/n}$ argue

$$\prod_1^{m-1} (1 - \frac{j}{n}) \sim e^{-m^2/2n}.$$

If $m = \sqrt{2n \log 2}$ then this is close to half. What is the relevance of this calculation with $m = 30$ and $n = 365$ to the birthday problem? Calculate this value $e^{-m^2/2n}$ for $m = 30, 35, 40$ and $n = 365$ and compare with previous answers to see how good is this approximation.

66. Let $n > 4$. Show that there exists a two colouring of the **edges** of the complete graph K_n so that the total number of monochromatic copies of K_4 is at most $\binom{n}{4} 2^{-5}$.

Lectures which really teach will never be popular; lectures which are popular will never teach. — V I Arnold calls this Faraday effect.

Clearness and truth are in quantum complementarity relation. —Niels Bohr

Humanity is moving towards bureaucratization where an all-powerful bureaucratic apparatus will be suppressing everything alive and creative that still exists. — V A Rokhlin.

Human beings can be called Stellar Nuclear Waste. — Martin Rees, Astronomer Royal.

67. X, Y are independent rvs on a space. $X \sim P(\lambda)$ and $Y \sim P(\mu)$, find the conditional distribution of X given $X + Y = n$. $X \sim B(n, p)$ and $Y \sim B(m, p)$, Given $X + Y = k$, find the conditional distribution of X . $X \sim G(p)$ and $Y \sim G(p)$, Given $X + Y = n$, find the conditional distribution of X . In each case Verify $E(E(X|X + Y)) = E(X)$.
68. I have given a letter to Chirayu to post. The chances that he forgets to mail is 0.1. Given that he posted, the chances that the letter reaches the addressee post office is 0.9. Given that it reached that post office, the chances that the postman fails to deliver is 0.1.
Given that the addressee did not get the letter, (i) what is the probability that Chirayu forgot to mail? (ii) Chirayu mailed but the letter did not reach that PO? (iii) it reached the PO but was not delivered?
69. Consider placing 3 balls in 3 boxes (Maxwell-Boltzman). Let N be the number of occupied boxes. Let X_i be the number of balls in box i for $i = 1, 2, 3$. Calculate joint distribution of (N, X_3) ; of (X_1, X_2, X_3) . Find $E(X_3|N)$ and $E(N|X_3)$.
70. Consider a probability space (Ω, P) . Let L be the collection of all bounded random variables on the space (that is, just all bounded functions on Ω). X, Y, Z below are from L . show
(i) L is a linear space and $X \mapsto E(X)$ is a linear map on L to the real line and $|E(X)| \leq \sup |X|$. Show $X \geq 0$ implies $E(X) \geq 0$.
(ii) $E(aX + bZ|Y) = aE(X|Y) + bE(Z|Y)$ Thus $X \mapsto E(X|Y)$ is a linear map on L to L . Also $|E(X|Y)| \leq \sup |X|$. Show $X \geq 0$ implies that $E(X|Y) \geq 0$. Show if X is the constant function $X \equiv 7$ then $E(X|Y) \equiv 7$.
(iii) If X, Y are independent then $E(X|Y)$ is the constant random

variable c where $c = E(X)$.

(iii) If X is a function of Y , then $E(X | Y) = X$. For example $E(\sin Y | Y) = \sin Y$.

71. Calculate pgf of $B(n, p)$ random variable; $P(\lambda)$ variable; $G(p)$ variable. Use these to show once again, sum of independent Poisson variables is again Poisson. Show that sum of r independent variables, each $G(p)$ is $NB(r, p)$.
72. There are N tickets numbered $1, 2, \dots, N$
- (i) If n tickets are chosen and X is the smallest number on the tickets selected; Find distribution of X .
- (ii) Suppose tickets are drawn one by one with replacement until a repeat occurs and then stop. Let X be the number of draws. Show

$$P(X = k) = (k-1)! \binom{N}{k-1} \frac{k-1}{N^k}; \quad k = 2, 3, \dots, N+1$$

Verify these all add to one. Show

$$\begin{aligned} E(X) = 2 + \left(1 - \frac{1}{N}\right) + \left(1 - \frac{1}{N}\right) \left(1 - \frac{2}{N}\right) + \dots \\ + \left(1 - \frac{1}{N}\right) \left(1 - \frac{2}{N}\right) \dots \left(1 - \frac{N-1}{N}\right) \end{aligned}$$

73. I toss a coin, chance of heads p , $0 < p < 1$. If Heads up, I roll a die and give you as many rupees as the face that shows up. If tails up, I toss two dice and give you as many rupees as the sum of numbers shown. Calculate distribution of the money you earn.
74. from an Urn containing a white and b black balls we draw n balls SRSWOR. Calculate the probability of drawing at least one white ball by two methods and show equals:

$$1 - \frac{b(b-1) \dots (b-n+1)}{(a+b)(a+b-1) \dots (a+b-n+1)} \quad (\bullet)$$

also

$$\begin{aligned} \frac{a}{a+b} \left[1 + \frac{b}{a+b-1} + \frac{b(b-1)}{(a+b-1)(a+b-2)} + \dots \right. \\ \left. + \frac{b(b-1) \dots (b-n+2)}{(a+b-1)(a+b-2) \dots (a+b-n+1)} \right] \quad (\dagger) \end{aligned}$$

Thus you get an identity, that these two quantities $(\bullet), (\dagger)$ are equal.

75. Each of n urns contain a white and b black balls. One ball selected at random from first urn is transferred to the second, then one ball selected at random from second urn is transferred to the third, and so on and finally one ball is drawn from the last. Show that the probability it is white equals $a/(a+b)$.

Hint: If p_k is probability of drawing white ball from k -th urn, then

$$p_{k+1} = \frac{a+1}{a+b+1}p_k + \frac{a}{a+b+1}(1-p_k)$$

76. From an urn containing a white and b black balls SRSWOR of size $a+b$ is drawn – thus all the balls are drawn one by one. Show that the probability first ball is white is same as the probability that last ball is white. (Hint: Think of symmetry)
77. You have 101 fair coins and I have 100 fair coins. We toss all the coins. What are the chances that you will have more H than me? Show it is $1/2$. (Think of symmetry).

Last few problems are from J V Uspensky, tricky, not ugly.

We should salute our honest judges and learn how to think. We should also salute our recent CJs and learn how not to think.

One CJ put himself on the bench to decide a case filed against him. One CJ invented 'secret proof': A committee or our rulers give a sealed cover based on which your case is decided – none of us will be told what is in the cover. One CJ invented 'artful justice': Everyone knew that collecting money by our rulers, with details kept secret from all except themselves, was undemocratic. CJ waited for years till the act was done, money used and then declared it improper. Our rulers wanted to set up an investigating committee when their spying on us became public. The CJ said a committee setup by rulers will not appear impartial and so he set up – none of us ever heard anything more about it. One CJ invented a 'Masterstroke': He disliked verdict of a two judge bench. After one of them retired, the CJ set up a three judge bench, with that not-yet-retired judge and two others, to overturn the verdict, by a majority decision. Wow.

78. Consider distribution of cards in a Bridge game. For $1 \leq k \leq 4$ Let N_k be the event that north has at least k aces. Similarly, S_k, E_k, W_k denote events for South, east and west. discuss the number of aces in west's possession in the following events:

(i) W_1^c . (ii) N_2S_2 . (iii) $N_1^cS_1^cE_1^c$. (iv) $W_2 \setminus W_3$. (v) $N_1S_1E_1W_1$. (vi) N_3W_1 . (vii) $(N_2 \cup S_2)E_2$.

Verify:

$S_3 \subset S_2$. (ii) $S_3W_2 = \emptyset$. (iii) $N_2S_1E_1W_1 = \emptyset$. (iv) $N_2S_2 \subset W_1^c$. (v) $W_4 = N_1^cS_1^cE_1^c$.

79. Verify: (i) $(A \cup B)^c = A^cB^c$ (ii) $A^c \cup B^c = (AB)^c$
(iii) $(A \cup B) - AB = AB^c \cup A^cB$ (iv) $(A \cup B)C = AC \cup BC$

80. State which of the following are correct.

(i) $(A \cup B) \setminus C = A \cup (B \setminus C)$ (ii) $ABC = AB(C \cup B)$
(iii) $A \cup B \cup C = A \cup (B \setminus AB) \cup (C \setminus AC)$ (iv) $(A \cup B) \setminus A = B$
(v) $(A \cup B)^cC = C \setminus C(A \cup B)$.

81. Let A, B, C be events. Find expressions for the following.

(i) Only A occurs (ii) both A and B occur but not C . (iii) At least one of them occurs. (iv) Two of them and no more occur (v) Not more than two of them occur. (vi) one of them and no more occurs.

82. Urn 1 contains 2 white and 1 black ball. Urn 2 contains 1 white and 5 blackballs. One ball is transferred from urn 1 to urn 2. Then a ball is drawn from urn 2 and this happens to be white. What is the probability that the transferred ball was black?
83. from an urn containing 3 white and 5 black balls, 4 balls are transferred to an empty urn. from this urn two balls are drawn and they both happen to be white. What is the probability that third ball taken from this urn will also be white when (i) the two drawn balls are returned (ii) when the two balls drawn are not returned.
84. From an urn containing 5 white and 5 black balls 5 balls are transferred to an empty second urn. From this three balls are transferred to an empty third urn. One balls is drawn from the third urn now and it is white. What is the probability that the five balls transferred from the first urn are white?
85. A box contains 2^n tickets among which $\binom{n}{k}$ tickets bear the number k ; for $0 \leq k \leq n$. A set of m tickets is drawn at random and S is the sum of the numbers on those tickets. Show

$$E(S) = \frac{1}{2}mn \quad \sigma_S^2 = \frac{1}{4}mn - \frac{m(m-1)n}{4(2^n - 1)}$$

86. A box contains k varieties of objects, the number of objects of each variety being the same. These objects are drawn one by one with replacement. Let N be the smallest number of drawings which produces objects of all varieties. Show (I have not checked)

$$E(N) = k \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k} \right)$$

$$\sigma_N^2 = k^2 \left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{k^2} \right) - k \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k} \right)$$

87. Consider coin tossing (chance of Heads p) with n -th toss giving $X_n = +1$ if heads and -1 if tails. Put $S_0 = 0$ and $S_n = \sum_1^n X_i$. Want to know when we hit 1 for the first time, that is

$$a_0 = 0; \quad a_1 = p; \quad a_n = P(S_n = 1; S_j \neq 1 \forall j < n) \quad n \geq 1$$

The event $(S_n = 1; S_j \neq 1 \forall j < n)$ is referred to as ‘first visit to one occurs at time $n + 1$ ’. In physics it is referred to as ‘first passage

through one occurs at time $n + 1$ '. We put $a_0 = 0$. Clearly $a_1 = p$. Let $\varphi(s) = \sum a_n s^n$ be the generating function of the sequence of numbers (a_n) . This is not pgf of any random variable at this stage. Show

$$a_n = q(a_1 a_{n-2} + a_2 a_{n-3} + \cdots + a_{n-2} a_1) \quad n > 1$$

Show that $\varphi(t) - pt = qt\varphi^2(t)$. Conclude that

$$\varphi(t) = \frac{1 - \sqrt{1 - 4pqt^2}}{2qt}; \quad a_{2k} = 0; \quad a_{2k-1} = \frac{(-1)^{k-1}}{2q} \binom{1/2}{k} (4pq)^k.$$

Conclude: $\varphi(1) = (1 - |p - q|)/(2q)$ which equals p/q if $p < q$ and equals one if $p \geq q$. Conclude that if $p < q$ then there is a positive chance of never reaching one.

Show $\varphi'(1) = 1/(p - q)$ if $p > q$ and equals ∞ when $p = q = 1/2$. Thus in fair coin tossing you surely reach one, but the expected waiting time is infinity. (why did I not consider $p < q$?)

88. In a branching process assume that the progeny generating function is $\varphi(t) = at^2 + bt + c$ where a, b, c are positive constants and $a + b + c = 1$. If the probability of extinction is e , where $0 < e < 1$ then show that $e = c/a$.

89. (Optional, essentially Analysis)

For a non-negative rv X values $(x_i, i \geq 1)$ with probabilities $(p_i, i \geq 1)$ expectation is meaningful, in the sense $\sum x_i p_i$ is either finite or infinity. Of course when it is ∞ , we decided not to define expectation. Temporarily, for this exercise, assume that in the later case we do define expectation to be infinity. Show that indeed, in all cases — that is, when the expectation is finite or not — the limit $[\varphi(1) - \varphi(t)]/[1 - t]$ exists and equals $E(X)$ and it equals $\sup\{\varphi'(t) : 0 < t < 1\}$.

(Most of the problems are from Feller and Uspensky)

Campbell's Law (Goodhart's Law): "when a measure becomes a target, it ceases to be a good measure" — people will manipulate a metric once it becomes a performance goal, causing it to lose its original value as a reliable indicator.

Sales: setting a sales target might lead employees to focus on increasing sales by unethical behaviour or neglecting service to undermine the original goal of sustainable growth. Software: A team might focus on increasing their software deployment without considering code quality, leading to more bugs and technical debt. Healthcare: A hospital trying to lower its patient length of stay might prematurely discharge patients, leading to more readmissions. Cobra Effect: A bounty on cobras led to people breeding them to collect the bounty, which increased the number of cobras rather than decreasing them. Nail Production: When the goal was to produce the highest number of nails, factories made millions of tiny nails. When the target shifted to weight, they made a few huge, impractical nails instead.

90. (These are problems in Analysis)
 - If $X \sim \text{Unif}(0, 1)$ find all moments of X . Do the same if $X \sim \text{Unif}(a, b)$.
 - If $X \sim N(0, 1)$, find all moments of X .
 - If $X \sim N(\mu, \sigma^2)$ show $E(|X - \mu|) = \sqrt{2/\pi} \sigma$.
 - If $X \sim \exp(\lambda)$ find all moments of X . Do the same if X is double exponential.
 - If $X \sim \beta(m, n)$ where $m \geq 1, n \geq 1$ are integers, find all its moments.
91. If f and g are pdfs and $0 < p < 1$ then $h(x) = pf(x) + (1 - p)g(x)$ is a pdf. Show this.

I toss a coin with chance of heads p . If heads, I select a number at random from $(0, 1)$ and if Tails, I select a number at random from $(4, 5)$. Find pdf of the number X so selected.
92. A number is selected at random from $(0, 1)$ and an equilateral triangle is constructed with this side. Find the pdf of its area
93. If $X \sim \text{unif}(-1, 1)$ find pdfs of $U = \sin(\pi X/2)$; $V = \cos(\pi X/2)$, $W = |X|$ and $Z = 2X \bmod (1)$.
94. Suppose that the radius of a sphere is a rv — due to inaccuracies of the manufacturing processes, the radii of different spheres may slightly

vary — with density $f(r) = 6r(1 - r)$ for $0 < r < 1$. Find the pdfs of Volume V and surface area S .

95. A fluctuating current I may be considered as a uniformly distributed rv over the interval $(9, 11)$. If the current flows through a 2-Ohm resistor, find the pdf of the power $P = 2I^2$.
96. The speed of a molecule in a uniform gas at equilibrium is a rv with pdf

$$f(v) = Cv^2e^{-bv^2} \quad v > 0$$

Here b is a constant given by $b = m/(2\kappa T)$ where κ is Boltzman constant, T is absolute temperature and m is mass of the molecule. Evaluate the constant C . Find the pdf of Kinetic energy of the molecule $E = mV^2/2$.

97. I have a random variable with moments μ_k , that is, $\mu_k = E(X^k)$ for $k \geq 0$, where, of course, $\mu_0 = 1$. Show that the following matrix is non-negative definite.

$$\begin{pmatrix} \mu_0 & \mu_1 & \cdots & \mu_k \\ \mu_1 & \mu_2 & \cdots & \mu_{k+1} \\ \mu_2 & \mu_3 & \cdots & \mu_{k+2} \\ \cdot & \cdot & \cdots & \cdot \\ \cdot & \cdot & \cdots & \cdot \\ \cdot & \cdot & \cdots & \cdot \\ \mu_k & \mu_{k+1} & \cdots & \mu_{2k} \end{pmatrix}.$$

98. Suppose that a random variable X has density f . Median of X is any number c such that

$$P(X \leq c) = \frac{1}{2} = P(X \geq c); \quad (i.e.) \quad \int_{-\infty}^c f(x)dx = \frac{1}{2}$$

Show that a median always exists.

Assume that X has a unique median m . For any real number b , show — assume needed integrals exist — that

$$E(|X - b|) = E(|X - m|) + 2 \int_m^b (b - x) f(x) dx$$

For what b is $E(|X - b|)$ minimum. What if median is not unique?

[Some problems are from P L Meyers Introduction to probability ...]

May All Be Happy ♦ May All Be Free From Illness.
 May All See The Auspicious ♦ May No One Suffer.
 Peace! Peace! Peace!

Upanishads

May the winds blow sweetly ✱
 May the streams scatter sweetness ✱
 May the herbs bring us sweetness ✱
 May nights and dawns fill us with sweetness ✱
 May the dust of earth be full of sweetness ✱
 May the heaven itself, like father, bring us sweetness ✱
 May the ancient forest trees be full of sweetness ✱
 May the Sun himself shower on us sweetness ✱
 May sweetness flow towards us from all directions.

Upanishads

99. X, Y have joint density $f(x, y) = 6(1 - x - y)$ for $0 < x < 1; 0 < y < 1; 0 < x + y < 1$. Calculate $P(2X + 3Y < 1)$ and $E(XY + 2X^2)$
100. X_1, X_2 have joint pdf $h(x_1, x_2) = 8x_1x_2$ for $0 < x_1 < x_2 < 1$. Find the joint density of $Y_1 = X_1/X_2, Y_2 = X_2$. Hence calculate the pdf of X_1/X_2 .
101. X, Y have joint pdf $f(x, y) = 2e^{-(x+y)}$ for $0 < x < y < \infty$. Find $E(Y | X = x)$.
102. Let X be unif $(0, 1)$. Find a function of X which is $B(n, p)$. Find a function of X which is Geometric. More generally, given any discrete random variable W , find a function of X which has the same distribution as W . If Z is Unif $(-10, 100)$ find a function of X which has the same distribution as Z .
- Define for each $n \geq 1$, X_n to be n -th decimal digit of X .
 (More precisely: remember X is a function defined on a probability space (Ω, P) , just that we decided not to see this space. Now for any sample point ω we define $X_n(\omega)$ as the n -th decimal digit of the number $X(\omega)$)
 Show that $X_n, n \geq 1$ is a sequence of iid rvs.
103. Let X, Y be independent uniform $(0, c)$. Find the conditional density and conditional expectation of X given $X + Y = z$. Do the same when they are $\exp(\lambda)$. Do the same when they are Gamma variables with parameters λ and μ .

104. For a sample of size two from $\text{Unif}(0,1)$, find the distribution of the sample mean, distribution of the geometric mean, and distribution of the standard deviation. (understand the problem first)
105. Let X and Y be independent uniform $(0, a)$. Show that

$$E|X - Y|^n = \frac{2a^n}{(n+1)(n+2)}$$

106. Joint pdf of (X, Y) is given by $f(x, y) = 2/[1 + x + y]^3$ for $0 < x < \infty, 0 < y < \infty$.
 Compute marginal density of X ; conditional density of Y given $X = x$.
 For fixed x compute $(1 + x + Y | X = x)$. compute $E(Y | X = x)$.
107. Suppose that (X, Y) is bivariate normal with $E(X) = E(Y)$ and $E(X^2) = E(Y^2)$. Show that $X + Y$ and $X - Y$ are independent.
108. The exponent of a bivariate normal is

$$-\frac{1}{6}[4(x+1)^2 - 2(x+1)(y-2) + (y-2)^2]$$

Find the means, variances, and covariance of the variates.

(Some of the problems are from R V Hogg and A T Craig)

The neurons of your brain show vigorous rhythmical activity when you are asleep – perhaps replaying what you learned that day. Loss of sleep hurts attention, executive functions, working memory, mood, quantitative skills, logical reasoning and even motor dexterity. When people become sleep deprived, their body's ability to utilize the food they are consuming falls by about one third. The ability to make insulin and to extract energy from the brain's favourite source, glucose, begins to fail miserably. At the same time you find marked need to have more of it because the body's stress hormone levels begin to rise in an increasingly deregulated fashion.

John Medina.

(Many problems are simple. Please think before you open your pen)

109. $(X_i, 1 \leq i \leq 5)$ are i.i.d. $\text{uniform}(0, 1)$. Let $(Y_i, 1 \leq i \leq 5)$ be the order statistic; that is, X_i arranged in increasing order. For example if $X_1 = 0.5, X_2 = 0.3, X_3 = 0.7, X_4 = 0.1, X_5 = 0.6$ then $Y_1 = 0.1, Y_2 = 0.3, Y_3 = 0.5, Y_4 = 0.6, Y_5 = 0.7$. Find the pdf of $(Y_i, 1 \leq i \leq 5)$.
110. Let $X_i, 1 \leq i \leq 5$, be $\exp(1)$ i.i.d variables and $Y_1 < \dots < Y_5$ be the order statistic. Show that Y_2 and $Y_4 - Y_2$ are independent.
111. Let $Y_1 < Y_2 < Y_3$ be an order statistic of size 3 from the distribution with density $f(x) = 2x$ for $0 < x < 1$. Show that $Y_1/Y_2, Y_2/Y_3$ and Y_3 are independent.
112. A three dimensional random vector (X, Y, Z) has p.d.f 6 in the tetrahedron with vertices $(0, 0, 0), (0, 0, 1), (0, 1, 0)$, and $(1, 0, 0)$. Find the p.d.f of X given Y .
113. Verify the following for $n = 2$ and $n = 3$. Let $X_1, \dots, X_n$ be iid $\text{uniform}(0, 1)$ and S_n is their sum. Then its density is given by

$$f_n(x) = \frac{1}{(n-1)!} [x^{n-1} - \binom{n}{1}(x-1)_+^{n-1} + \binom{n}{2}(x-2)_+^{n-1} - \dots]$$

for $0 < x < n$. Here a_+ denotes a or 0 according as $a > 0$ or not.

f_1 is discontinuous at some points. f_2 is continuous everywhere but not differentiable at some points. f_3 is differentiable everywhere but second derivative does not exist at some points.

114. Let X, Y and Z be independent uniform $(0, 1)$ and $S = X + Y + Z$. Find the conditional density of S given $Y = y$ and $Z = z$. Find the conditional density of (Y, Z) given $S = s$.
115. Let X, Y be independent uniform $(0, 1)$. Find the joint density of U, V where $U = X + Y$ and $V = X - Y$. Find density of $X + Y$. Find density of $X - Y$. Sketch their graphs. Find density of $Z = XY$. Find density of $W = X^{85}$.
116. Suppose that the lifetime, in hours, of a radio tube can be modelled as continuous random variable X with density $f(x) = 100/x^2$ for $x > 100$. What is the conditional probability that a tube will last more than 200 hours if it is still functioning after 150 hours of service. What is the probability that if three such tubes are installed in a set, exactly one will have to be replaced after 150 hours of service. What is the maximum number of tubes that may be inserted into a set so that there is a probability of at least 0.25 that after 150 hours of service all of them are still functioning?
117. The speed of a molecule in a uniform gas at equilibrium is a random variable V with pdf given by

$$f(v) = a v^2 e^{-bv^2} \quad \text{for } v > 0$$

where $b = m/2kT$. Here k is Boltzman constant, m is the mass of the molecule and T is the absolute temperature. Evaluate the constant a in terms of b .

Derive the distribution of the kinetic energy $W = mV^2/2$.

118. Let $(X_i, 1 \leq i \leq n)$ be iid uniform $(0, 1)$. Show that the product $Y = \prod_1^n X_i$ has density $f(t) = \frac{1}{n-1} [\log(1/t)]^{n-1}$ for $0 < t < 1$. Hint: deal with $-\log X_i$. Suppose we have colliding granules of a mineral. Concentrate on one granule of unit mass. After one collision it breaks into two and concentrate on a randomly selected part. This piece after one collision breaks into two and concentrate on one randomly selected part and so on. What will be the mass of the selected piece after n collisions? We assume that the collisions are independent and after a collision a mass m is reduced to mX , where X uniform $(0, 1)$. This process is also called splitting process.
119. We have two envelopes one containing a rupees and b rupees; $a \neq b$ strictly positive integers. You pick one of the two envelopes at random

and see what it has. Is it possible to say, with probability larger than $1/2$ whether the amount you have seen is smaller or larger of the two amounts?

C. M. Greenstead and J. L. Snell say yes. Here is how. Toss a fair coin till a head turns up and let Z be the number of tosses needed plus $1/2$. If what you have seen is smaller than Z , say 'you have seen the smaller'; If what you have seen is larger than Z , say 'you have seen the larger'. Check that this works. Does it sound funny?

A glimpse of contemporary IKS —

We refused drinking straw for an eighty year old prisoner who was unable to manage. We beat to death a mentally retarded person for partaking sweets at a temple, as he was of different sect. We threw an English lecturer, paraplegic from birth, in jail for nearly a decade and then released — no evidence of wrong doing. We blindfolded our own citizens, including a pregnant mother and little children, and pushed into a neighbouring country. Our Emperor incites people of one state against another (for example, of Bihar against Tamilnadu/Karnataka) by false narratives. We close down a medical college when favoured group is not in merit list. Lynching, harassing, hate speeches, destroying educational institutions and religious places ... are birth right of a favoured group. These show:

For Honourable first citizen of the country – We are heading to RamRajya.

For courts, rulers and media — tiny insignificant forgettable news items.

For me — A knowledge of our Character.

120. (i) (Memoryless property) Suppose $X \sim \exp(\lambda)$. Show that $P(X > a + b | X > a) = P(X > b)$ for $a > 0$ and $b > 0$.
 (ii) Darshan, Karshit and Rhyin enter a service station (post office or ATM etc) where there are two counters, both vacant. D and K stand at the two counters. R is served as soon as a counter is free. Assume that the service times of the three are independent exponential with parameter λ and no time is lost between different services. Do you think R would be the last to leave the station?
121. Define
 $F(x) = 0$ for $x < 21$; $F(x) = 0.2$ for $21 \leq x \leq 23$; $F(x) = 0.2 + \frac{x-23}{4}$ for $23 \leq x \leq 25$; $F(x) = 0.7$ for $25 \leq x < 345$ $F(x) = 1$ for $x \geq 345$.
 If $X \sim F$, Find $P(21 < X < 22)$ $P(21 \leq X \leq 22)$
 $P(22 \leq X \leq 24)$ $P(24 \leq X \leq 345)$.
122. I toss a coin whose chance of heads is 0.3. If heads up I pick a number at random from the interval $(0, 3)$. If tails up I pick a number at random from the interval $(2, 4)$. Let X be the number. Find distribution function of X . Does it have a density?
 Do the same thing for the following. I toss a coin, chance of heads is 0.3. If heads up, I pick a number at random from $[0, 4\pi]$. If tails up, I pick a number at random from among the set of points in the same interval where the sine function is zero. Let X be the number.

123. I pick a point at random from $(0, 1)$, say X . I put $Y = X$. Do you think (X, Y) has joint density? Justify.
124. The proportion of Alcohol in a certain compound can be thought of as a continuous random variable X with density $f(x) = 20x^3(1-x)$ for $0 < x < 1$. Obtain the df F of X and sketch it. Find $P(X \leq 2/3)$.
Suppose that the selling price of the compound depends on the alcohol content as follows : If $1/3 < X < 2/3$, then it sells for α rupees per gallon; otherwise it sells for β rupees per gallon. If the cost price is γ rupees per gallon find the probability distribution of the net profit per gallon.
125. The diameter of an electric cable is assumed to be a random variable X with density $f(x) = 6x(1-x)$ for $0 \leq x \leq 1$. Sketch the pdf. Sketch the df.
126. If the random variable K is uniformly distributed over $(0, 5)$, what is the probability that the roots of the following equation are real?
 $4x^2 + 4xK + K + 2 = 0$
127. (Two parameter Cauchy) Fix $\sigma > 0$ and $\mu \in R$. $f(x) = \frac{\sigma}{\pi} \frac{1}{\sigma^2 + (x-\mu)^2}$ for $x \in R$ Calculate its cdf. Does this have expectation?
128. (Laplace/two sided exponential). Fix $\lambda > 0$ and $\mu \in R$. $f(x) = \frac{1}{2}\lambda e^{-\lambda|x-\mu|}$ for $x \in R$. Calculate its df, expectation and variance.
129. (lognormal). Fix $\sigma > 0$; $\mu \in R$, $f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-(\log x - \mu)^2 / 2\sigma^2}$; for $x > 0$. Show that if $Y \sim N(\mu, \sigma^2)$ then $X = e^Y$ has the above density.
130. (logistic). Fix $a, b > 0$. Show $F(x) = \frac{1}{1+e^{-(a+bx)}}$ for $x \in R$ is a df. Calculate pdf.
131. If $X \sim N_k(\mu, \Sigma)$ and $a \in R^k$, $a \neq 0$. Show $Z_a = a^t X$ is univariate normal. Calculate its mean and variance. Suppose $a, b \in R^k$ are two linearly independent vectors. Show (Z_a, Z_b) is bivariate normal. Calculate its mean vector and covariance matrix.
132. Is it possible to load a die so that when you toss twice, all 11 scores are equally likely?

Speak gently! – It is better far	◆ to rule by love, than fear –
Speak gently – let not harsh words mar	◆ the good we might do here!
Speak gently! – Love doth whisper low	◆ the vows that true hearts bind;
And gently Friendship's accents flow;	◆ affection's voice is kind.
Speak gently to the little child!	◆ Its love be sure to gain;
Teach it in accents soft and mild: –	◆ it may not long remain.
Speak gently to the young, for they	◆ will have enough to bear –
Pass through this life as best they may,	◆ 't is full of anxious care!
Speak gently to the aged one,	◆ grieve not the care-worn heart;
The sands of life are nearly run,	◆ let such in peace depart!
Speak gently, kindly, to the poor;	◆ let no harsh tone be heard;
They have enough they must endure,	◆ without an unkind word!
Speak gently to the erring – know,	◆ they may have toiled in vain;
Perchance unkindness made them so;	◆ oh, win them back again!
Speak gently! – 't is a little thing	◆ dropped in the heart's deep well;
The good, the joy, which it may bring,	◆ eternity shall tell. (by David Bates)

133. Random Walk is an **Idea**. Here is some Alap. I toss a coin whose chance of heads in a single toss is p , where $0 < p < 1$. Heads up, I move one step forward; tails up I move one step backward, as suggested further below. Write down the transition probabilities in the following cases. In each case calculate distribution of X_2 after choosing some initial starting point X_0 .
- (a) The state space is all integers. This is usual RW or **unrestricted RW**. We did in class with $p = 1/2$.
 - (b) The state space is $\{0, 1, 2, \dots\}$. If you are at 0 then move to 1. This is RW **reflected** at zero. Imagine mirror placed at $1/2$.
 - (c) State space is $\{0, 1, 2, \dots, 99\}$. If at 0, move to 1. If at 99, move to 98. This is RW with two **reflecting barriers**.
 - (d) The state space is as in (c). If at 0 or at 99, do not move; stay there forever. This is RW with two **absorbing barriers**.
 - (e) State space is as in (c). If at 0, move to 1. If at 99, stay there. This is RW with **one absorbing barrier and one reflecting barrier**.
 - (f) State space is as in (c). Fix two numbers $0 < r_0, r_{99} < 1$. If at 0 stay there with probability r_0 and go to 1 with probability $1 - r_0$. Similarly do at 99 using r_{99} . This is RW with two **elastic barriers**.
 - (g) State space is as in (c). Treat 0 as state next to 99 and 99 as state preceding 0; (i.e.) think of the states arranged in a circular/cyclic order; (i.e.) do addition modulo 100. This is **cyclic RW**.
134. (Gambler's Ruin) consider random walk with two absorbing barriers 0, N . state space is $\{0, 1, 2, \dots, N\}$ and the transition mechanism is: win

a rupee with probability p or lose a rupee with probability $1 - p = q$; (i.e.) From i go to $i + 1$ or $i - 1$ with these probabilities.

Let u_i = probability of getting absorbed at N starting from i .

Argue: $u_0 = 0, u_N = 1$. Find a recurrence relation for $\{u_i\}$. Show

$$u_i = \frac{1 - [q/p]^i}{1 - [q/p]^N} \text{ when } p \neq 1/2; \quad u_i = i/N \text{ if } p = 1/2.$$

135. State space consists of the $52!$ arrangements of the usual deck of playing cards. Here are some laws of motion. In each case explain your vocabulary and then the transition matrix:
- (A) interchange bottom card with a randomly selected card.
 - (B) Select a card at random and put it on top.
 - (C) select a card at random, interchange its neighbours (think cyclic, top card and bottom card are neighbours).
 - (D) Pick two cards (with replacement) at random and interchange them. What if it is without replacement.
 - (E) Cut at random and interchange the stacks. That is, pick a number at random from among $\{1, 2, \dots, 51\}$. The stack is now split into two parts: initially: top $[1, k]$ and bottom $[k+1, 52]$. New stack: new 1 = old $(k+1)$; new 2 = old $(k+2)$; new $(52 - k) = \text{old } 52$; and new $(52 - k + 1) = \text{old } 1$, etc new 52 = old k . (You can interlace the two parts, a whole theory lies ahead. Persi Diaconis is an expert.)
136. A die is consecutively turned from one face to any of the four neighbouring faces with equal probability and independent of the preceding turns. Find $\lim_n p_{66}^{(n)}$.
137. Let (V, G) be a finite graph. Here is random walk on V . If at a vertex v move to one of its neighbours at random. if there are no neighbours, do not move.
When is a vertex absorbing? when is it irreducible?
Assume it is irreducible. If it is bipartite, can it be aperiodic? must the period be two?
138. State space consists of the 64 squares of a chess board. There is only one piece on the board, say, a knight on one square. From a state, you select one of its possible Knight moves at random and move there. Is this irreducible chain? What is the period? What is the mean time taken by the knight to return to the starting position? Do the same thing with another chess piece of your choice.

You go and oppose the prevailing faith, you go and criticise a hero, a great-man, who is generally believed to be above criticism because he is thought to be infallible, the strength of your argument shall force the multitude to decry you as vainglorious. This is due to the mental stagnation. Criticism and independent thinking are the two indispensable qualities of a revolutionary.

Bhagat Singh.

Supreme court Gems: 1 When a learned judge gave a dissenting opinion, they did not make it public saying 'if the dissent has merit then the other judges would have taken note of it'! 2 'Why should anyone be scared of Government surveillance? If you have nothing to hide, why should you be afraid' !! 3. Why are you worried if your name is removed from voter list; Let elections be over, we shall reinstate your name, if valid!!!

139. some results, explained but not completely written down in class. write their proofs. Make sure, you do not use intuition as justification.

$(X_n, n \geq 0)$ is a MC with transition matrix P .

$$P(X_{15} = i_{15} | X_0 = i_0, X_2 = i_2, X_4 = i_4, \dots, X_{10} = i_{10}, X_{11} = i_{11}) = P(X_{15} = i_{15} | X_{11} = i_{11}) = P(X_4 = i_{15} | X_0 = i_{11})$$

equal to (i_{11}, i_{15}) element of the matrix P^4 .

Let $1 \leq m_1 < m_2 < m_3 < m_4 < n$. Then

$$P(X_n = j | X_{m_1} = i_1, X_{m_2} = i_2) = P(X_n = j | X_{m_2} = i_2) = P(X_{n-m_2} = j | X_0 = i_2).$$

$$P(X_n = j | X_{m_1} = i_1, X_{m_2} = i_2, X_{m_3} = i_3, X_{m_4} = i_4) = P(X_n = j | X_{m_4} = i_4) = P(X_{n-m_4} = j | X_0 = i_4)$$

Feel in words what these equations are telling you.

140. Consider a MC with state space S and transition matrix P . For $i, j \in S$, Define $i \rightsquigarrow j$ (read: i leads to j) if $p_{ij}^{(n)} > 0$ for some $n \geq 1$. Define $i \longleftrightarrow j$ (read: i communicates with j) if $i \rightsquigarrow j$ and $j \rightsquigarrow i$. Define $i \sim j$ if $i = j$ or $i \longleftrightarrow j$.

(i) Show $i \rightsquigarrow j$ and $j \rightsquigarrow k$ implies $i \rightsquigarrow k$.

(ii) Give an example of a chain to show $i \rightsquigarrow i$ need not be true.

(iii) Show $\sim$ is an equivalence relation on S . The equivalence classes are called communicating classes.

A state i is called recurrent if $P(X_n = i \text{ for some } n \geq 1 | X_0 = i) = 1$.

(iv) Assuming that S is finite show that: If $i \rightsquigarrow j$ implies $j \rightsquigarrow i$ then i is recurrent.

(v) If i is recurrent then show $i \rightsquigarrow j$ implies $j \rightsquigarrow i$.

A state i is called **transient** if $P(X_n = i \text{ for some } n \geq 1 | X_0 = i) < 1$.

(vi) Assume S is finite. If i is transient, then show that there is j such that $i \rightsquigarrow j$ but $j \not\rightsquigarrow i$. Considering Z^3 argue that this statement may be false if S is infinite.

(vii) Show that in a communicating class, either every state is recurrent or every state is transient.

(viii) In a communicating class, show that every state has the same period.

(ix) A state i is called **absorbing** if $p_{ii} = 1$. Show that if $j \neq i$, $j \rightsquigarrow i$ and i is absorbing, then j must be transient.

(x) explain communicating classes/transient, recurrent, absorbing states in the following chains.

$$(a) \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}; (b) \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 1/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

(c) State space $S = \{0, 1, 2, 3, \dots\}$. $p_{0i} = 1/2^{i+1}$ and $p_{i,i-1} = 1$ for $i > 0$.

(d) State space $S = S_n$ permutation group on n element set, say, $\{1, 2, \dots, n\}$. Fix two permutations, η, ξ . Here is transition mechanism: $p_{\pi, \pi\eta} = p_{\pi, \pi\xi} = 1/2$.

141. For discrete random variables we have discussed mgf (moment generating function). Recall $M_X(t) = E(e^{tX})$. Thus if X takes the finite or countably many values $(x_i; i \geq 1)$ with corresponding probabilities $(p_i, i \geq 1)$ then $M_X(t) = \sum e^{tx_i} p_i$. Of course, this may not be finite for all t . So it is defined only for those values of t for which this is finite.

For a rv X with density f we can define its mgf by the formula $M_X(t) = E(e^{tX}) = \int e^{tx} f(x) dx$. Again this may not be defined for all $t \in R$.

Explaining the domain, calculate mgf of the rvs:

$$\text{uniform}(a, b) \quad \exp(\lambda) \quad N(0, 1) \quad N(\mu, \sigma^2).$$

The amount of time you spend tells you if you are understanding or not. Generally most of the problems are routine, many are simple. Yes, few are either boring or need work. If everything were trivial, the course is not worth it!

Democracy is now a Fantasy for us.

◆ The EC tells us: All citizens of India are guilty until they can prove themselves otherwise; following the rules and time frame set by me.

◆ The Supreme Court tells us: we restore your voting rights after the elections; If you criticize us, we banish you without hearing you.

◆ Rulers tell us: vote for double engine government, otherwise we do not care for you. Double engine appears to me unethical, illegal, scandalous. Once I asked, Honourable Justice Sri Alladi Kuppaswamy, a foolish question: Does a winner represent only those voted for him/her or all people of the constituency. His unequivocal answer was: he/she is bound to represent all people in the constituency.

142. Show that top-to-random shuffle is irreducible. Verify its stationary distribution is uniform.
143. Show Ehrenfests model has period two and its invariant distribution is Binomial $1/2$.
144. Describe communicating classes, Classify the states, of the following chains whose transition matrices are given below.

$$\begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 \\ 1/4 & 1/2 & 1/2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1/2 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 0 & 0 & 1/3 & 1/3 & 1/3 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 1/4 & 3/4 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1/3 & 2/3 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1/2 & 0 & 1/2 & 0 & 0 \\ 1/4 & 1/2 & 1/4 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 0 & 1/2 & 1/2 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1/4 & 0 & 1/2 & 0 & 0 & 1/4 \\ 0 & 1/4 & 0 & 1/2 & 0 & 1/4 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1/16 & 1/16 & 1/4 & 1/4 & 1/8 & 1/4 \end{pmatrix}$$

Incidentally, as far as discussing the nature of the states is concerned, is it necessary to have the full matrix before you? Instead of exact numbers in the above matrices I just put * when there is a strictly positive quantity. Would this have been enough for you?

145. This walk, called Burnside process, was invented by computer scientists L. Goldberg and M. Jerrum.

G is a finite group acting on a finite set X . This means for each $g \in G$ there is a map $x \mapsto g(x)$ of X such that:

- (i) $e(x) = x$ for all x , where e is identity element of the group; and
- (ii) $g_1(g_2(x)) = (g_1g_2)(x)$ for all $g_1, g_2 \in G$ and $x \in X$.

Here is a walk with state space X :

If at $x \in X$, pick a h at random from the set $G_x = \{g : g(x) = x\}$ and then pick a y at random from the set $\chi(h) = \{z : h(z) = z\}$. Move to y . This is one step of the walk on X .

Show that this is irreducible and aperiodic. If O_x is the G -orbit of x , that is, $O_x = \{g(x) : g \in G\}$ then show that transition matrix and stationary distribution are given by

$$p(x, y) = \frac{|O_x|}{|G|} \sum_{g \in G_x \cap G_y} \frac{1}{|\chi_g|}; \quad \pi(x) \propto \frac{1}{|O_x|}.$$

Conclude that limiting distribution restricted to orbits is uniform.

Special case: Let $k \geq 1$ and $n \geq 1$ be integers. Let $X = \{1, 2, \dots, k\}^n$; that is, all sequences $(i_1, i_2, \dots, i_n)$ where each $i_m \in \{1, 2, \dots, k\}$. This space represents arrangements of n (distinguishable) balls in k boxes; m -th coordinate tells the box where ball m is placed.

Let S_n be the group of permutations of $\{1, 2, \dots, n\}$. Then $\eta \in S_n$ acts on X by permuting coordinates: $\eta(i_1, i_2, \dots, i_n) = (i_{\eta(1)}, i_{\eta(2)}, \dots, i_{\eta(n)})$. Show that this is actually a group action.

Each orbit corresponds to an arrangement of n indistinguishable balls in k boxes. Convince yourself. Bose-Einstein proposal is precisely that these arrangements are equally likely. Thus this Markov chain implements B-E proposal and was extensively studied by Persi Diaconis.

146. Consider $N \times N$ chess board. I have large collection of stones each having one of k colours. Configurations, that is state space S , consist of placing of one stone in each of the squares. Here is how we walk from one configuration to other: pick a square s at random, and then pick one of its neighbours t at random. Replace stone at s by a stone of the colour in t .

Who are neighbours? State space is $\{(a, b); 0 \leq a, b \leq N - 1\}$.

(a, b) is neighbour of (c, d) if $[a = c \pm 1 \pmod N \text{ and } b = d]$ OR $[a = c \text{ and } b = d \pm 1 \pmod (N)]$; just like in random walk but with torus background.

For each set A of colours, let S_A be the set of all configurations using colours from A only. Show this set is closed, which means, if you are in S_A , you will stay in S_A forever.

Show that absorbing states are monochromatic configurations, that is each monochromatic configuration is a closed class and every configuration leads to some monochromatic configuration (and hence, by an earlier exercise, each non-monochromatic configuration is transient; after some stage you stop visiting it).

147. There is a list of k books. Each day, one of these k possible books are requested – the i -th one with probability $P_i(> 0)$. The list is revised as follows: The book selected today is moved to the top of the list, while the relative positions of all the other books remains unchanged. The state of the system is the ordered list of the books. How does the transition matrix look like? Discuss irreducibility and periodicity.

For any state η which is a permutation of $1, 2, \dots, k$ (understanding; $\eta(1)$ is the top name in the list, $\eta(2)$ is the second name in the list etc). let $\Pi(\eta)$ denote the limiting probability (stationary distribution). In order to be in this state η , it is necessary that the last request was for $\eta(1)$; the last non- $\eta(1)$ request be for $\eta(2)$; the last non- $\{\eta(1), \eta(2)\}$ request be for $\eta(3)$ etc. So it appears intuitively that

$$\Pi(\eta) = P_{\eta(1)} \frac{P_{\eta(2)}}{1 - P_{\eta(1)}} \frac{P_{\eta(3)}}{1 - P_{\eta(1)} - P_{\eta(2)}} \dots$$

Do with $k = 3, k = 4$, enough.

This is called Tsetlin Library, the Russian mathematician M L Tsetlin considered it first in 1963. This model very well applies to k -datasets instead of k books.

148. S set of all graphs, with at least one edge, on $V = \{1, 2, \dots, n\}$ is state space. Here is rule of motion. Start with a graph $G \in S$. Pick two vertices one after the other without replacement, say, u, v in that order. Disconnect v from all its neighbours and connect to u . Let $F \subset S$ be the set of all non-trivial (?) forests (disjoint union of trees). Show that if you start with an element of F , then you move only to an element of F . Show that every state leads to an element of F .

The freedom to doubt is an important matter in the sciences and, I believe, in other fields. It was born of a struggle. It was a struggle to be permitted to doubt, to be unsure. I feel a responsibility to proclaim the value of this freedom and to teach that doubt is not to be feared, but that it is to be welcomed as the possibility of a new potential for human beings. If you know that you are not sure, you have a chance to improve the situation. I want to demand this freedom for future generations. R Feynman

A University is a seat of learning, not a centre of worship. It believes in the pursuit of knowledge and not in the establishment of a cult. As university men it is our privilege and honour to seek for truth and in this pursuit we should not be deterred by the fear of what we might find.

S Radhakrishnan

This is strictly within our four walls, otherwise our rulers and courts will confine you to different four walls.

149. Let $Y_1 < Y_2 < Y_3$ be an order statistic of size 3 from the distribution with density $f(x) = 5x^4$ for $0 < x < 1$. Find the pdf of Y_3 .
150. $X_1, \dots, X_{23}$ is a sample of size 23 from a normal distribution. Find the joint distribution of $\sum_1^{15} X_i$ and $\sum_9^{21} X_i$.
151. X, Y, Z have the joint density

$$f(x, y, z) = \frac{6}{(1 + x + y + z)^4} \quad \text{for } x, y, z > 0$$

Find the distribution of $X + Y + Z$.

152. $F(x) = 0$ for $x \leq 0$ and $F(x) = x^2$ for $0 \leq x < 1/2$ and $F(x) = 1/2$ for $1/2 \leq x \leq 1$ and $F(x) = (x + 1)/4$ for $1 \leq x \leq 3$ and $F(x) = 1$ for $x \geq 1$. Show that this is a distribution function. Let X be a random variable with this DF. Calculate $P(1/2 < X < 1)$; $P(1/2 \leq x \leq 2)$; $P(X \geq 1/4)$.
153. Suppose X has density $f(x) = 1/2$ for $-1 \leq x \leq 0$ and $f(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ for $x > 0$. Calculate $P(X > -0.8)$.

154. Suppose X has DF $F(x) = 0$ for $x < 1$; $F(x) = x^4/4$ for $1 < x < 2$ and $F(x) = 1/2$ for $2 < x < 5$ and $F(x) = 1$ for $x > 5$. calculate values of F at $x = 1, 2, 5$. Find the probability that X is an integer. Find the probability $P(2 < X < 6)$ and $P(2 \leq X \leq 6)$.
155. Let $f(x, y) = 4xy$ for $0 < x < 1$ and $0 < y < 1$. Find $P(0 < X < 1/2; 1/4 < Y < 1)$ and $P(X < Y)$ and $P(X = Y)$ and $P(Y \leq X)$.
156. I have two random variables X, Y with some pdf. We know $P(X \leq 2, Y \leq 4) = 7/8$ and $P(X \leq 2, Y \leq 1) = 4/8$ and $P(X \leq 0, Y \leq 4) = 3/8$ and $P(X \leq 0, Y \leq 1) = 2/8$. Find $P(0 \leq X \leq 2, 1 \leq Y \leq 4)$.
157. If X, Y are uniform $(0, 1)$ find the density and DF of XY .
158. X, Y have joint pdf $f(x, y) = 6(1 - x - y)$ for $0 < x, y, x + y < 1$. Compute $P(2X + 3Y < 1)$.
159. X, Y have joint pdf $f(x, y) = 2e^{-x-y}$ for $0 < x < y < \infty$ Find the joint pdf of $(U, V) = (2X, Y - X)$
160. X, Y have joint pdf $f(x, y) = 8xy$ for $0 < x < y < 1$. Find the pdf of $(U, V) = (X/Y, Y)$.
161. X, Y have joint pdf $f(x + y) = x + y$ for $0 < x, y < 1$. Find the conditional density of Y given $X = a$. Hence calculate the conditional mean and conditional variance of Y given $X = a$.
162. Suppose a man leaves for work between 8 : 00 *a.m.* and 8 : 30 *a.m.* and takes between 40 to 50 minutes to get to office. Let X denote his time of departure and Y denote his travel time. Assume that they are independent and uniformly distributed. Find the probability that he reaches office before 9 : 00 *a.m.*.
163. If $X \sim N(\mu, \sigma^2)$ show $E(|X - \mu|) = \sqrt{2/\pi} \sigma$.

(some of these problems are standard and others are from R V Hogg and A Craig.)

